

JEE MAIN REVISION MASTER: PERMUTATIONS AND COMBINATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 218 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Fundamental Principles & Permutations (Distinct & Identical)	191 PYQs	Arrangements
Very High	Combinations, Selection & Multinomial Partitioning	28 PYQs	Selections & Distribution

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Fundamental Principles & Permutations (Distinct & Identical)	${}^n P_r = \frac{n!}{(n-r)!}, \quad \text{Arrangements with duplicates: } \frac{n!}{p! q! r!}, \quad \text{Circular: } (n-1)!$
Combinations, Selection & Multinomial Partitioning	${}^n C_r = \frac{n!}{r!(n-r)!}, \quad \text{Non-negative solutions: } {}^{n+r-1} C_{r-1}, \quad \text{Derangements: } D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Fundamental Principles & Permutations (Distinct & Identical) (Very High Yield — 191 PYQs)

Core Principle: Multiplication and addition principles, linear permutations with repetitions, rank of words in dictionary, and circular arrangements.

■ **Caution / JEE Pitfall:** For circular permutations of necklaces/garlands where clockwise and anticlockwise arrangements are indistinguishable, divide by 2: $((n-1)! / 2)$.

■ **Speed Trick:** Gap method: Place items that cannot be adjacent into the gaps created by the remaining unrestricted items.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A person has three different bags and four different books. The number of ways, in which he can put these books in the bags so that no bag i..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $A=(a,b,c):a,b,c$ are non-negative integers and $a+b+2c=22$. Then $n(A)$ is equal to ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The number of elements in the set $S=(r,k):k \in \mathbb{N} \text{ and } {}^{36}C_{(r+1)}=\frac{6({}^{35}C_r)(k^2-3)}{k}$ is ..." [Ans: B]

3.2 Combinations, Selection & Multinomial Partitioning (Very High Yield — 28 PYQs)

Core Principle: Combinatorial selection, Pascal identity ${}^nC_r + {}^nC_{(r-1)} = {}^{n+1}C_r$, distribution of identical items into distinct bins, and derangements.

■■ **Caution / JEE Pitfall:** Number of positive integral solutions ($x_i \geq 1$) to $x_1 + x_2 + \dots + x_r = n$ is ${}^{n-1}C_{(r-1)}$, whereas non-negative ($x_i \geq 0$) is ${}^{n+r-1}C_{(r-1)}$.

■ **Speed Trick:** Number of derangements of 4 items is $D_4 = 9$; for 5 items is $D_5 = 44$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The number of ways, in which 16 oranges can be distributed to four children such that each child gets at least one orange, is..." [Ans: N/A]
- **[2026 Morning Shift] Q#2 (Numerical):** "The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0,1,2,5,9, if the repet..." [Ans: 42]
- **[2026 Morning Shift] Q#3 (MCQ):** "The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters ..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

The number of elements in the set $S = \{(r, k) : k \in \mathbb{N} \text{ and } {}^{36}C_{(r+1)} = \frac{{}^{35}C_r}{k^2-3}\}$ is :

- (A) 2
- (B) 4 [Correct]
- (C) 8
- (D) 16

Correct Answer: B

Solution Analysis: We need to find the number of ordered pairs (r, k) satisfying ${}^{36}C_{(r+1)} = \frac{{}^{35}C_r}{k^2-3}$, where $k \in \mathbb{N}$. Let us simplify the relation using the standard NCERT combination formula. We know that ${}^nC_{(r+1)} = \frac{n!}{(r+1)!(n-r-1)!}$ and ${}^nC_r = \frac{n!}{r!(n-r)!}$. So, $\frac{{}^{36}C_{(r+1)}}{{}^{35}C_r} = \frac{(36! / (r+1)!(35-r-1)!)}{(35! / r!(35-r)!)} = \frac{(36! / 35!) \cdot (r! / (r+1)!)}{(35-r-1)! / (35-r)!} = \frac{36}{r+1} = \frac{36}{r+1}$. Hence, ${}^{36}C_{(r+1)} = {}^{35}C_r \cdot \frac{36}{r+1}$. Given, ${}^{36}C_{(r+1)} = \frac{{}^{35}C_r}{k^2-3}$. Comparing both expr

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#2)

The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0,1,2,5,9, if the repetition of the digits is allowed, is ____

Correct Answer: 42

Solution Analysis: To find numbers of number greater than 5000 and less than 9000 so number will start with 5 or 9 of the form 5abc and a,b,c $\in$ 0,1,2,5,9 also the number is divisible by 3. This means $5+a+b+c$ is divisible by 3 so the sum $a+b+c$ is of the type $3n+1$. unit place c determined by the sum of previous 3 places. $\text{rem}(a+b) = \text{remainder of } a+b$
 combination (a,b) value of c total = combination (a, b) $\times$ c(choice)
 0 (0, 0), (0, 9), (9, 0), (9, 9), (1, 2), (2, 1), (5, 1), (1, 5) 1 8
 $\times 1 = 8$ 1 (0, 1), (9, 1), (1, 0), (1, 9), (2, 2), (2, 5), (5, 2), (5, 5) 0, 9 $8 \times 2 = 16$ 2 (0, 2), (0, 5)